WEEK 10 • IS321 ENTERPRISE ARCHITECTURE

Linking EA and Technology Planning & Intelligent Enterprise Systems

The Technologies That Support the Business — and How AI Is Reshaping Them

TODAY'S ROADMAP

What We'll Cover

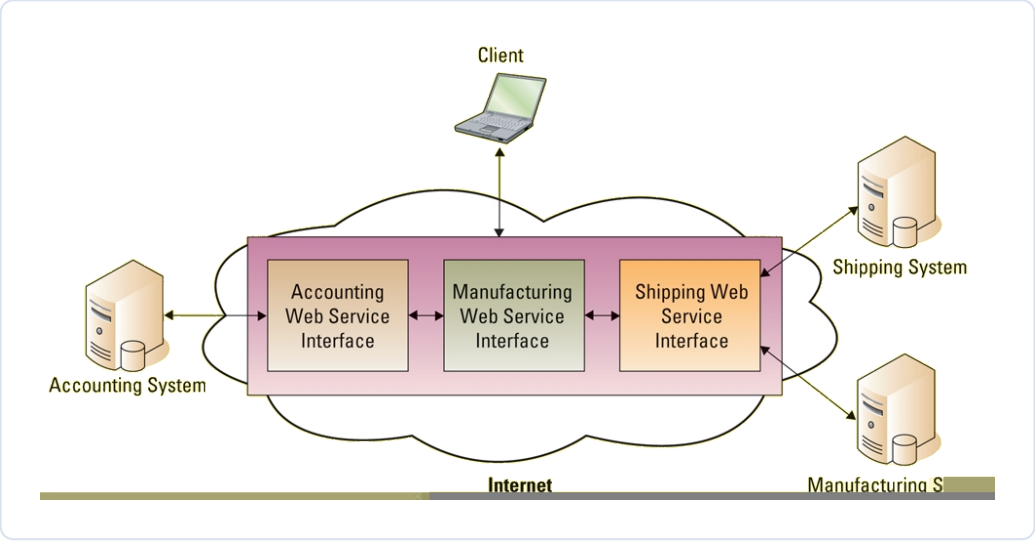
google drive + Gmail are physical hosted externally (3rd party service) on Google's infrastructure

→ university acts as a user that simply acquires these services to support their processes via Google Workspace for Education

01	**Web Services & Cloud Computing**	The NIST cloud model, and the IaaS/PaaS/SaaS service tiers
02	**Service-Oriented Architecture (SOA)**	Turning siloed applications into reusable business services
03	**Virtualization & Grid Computing**	Getting more out of existing hardware investments
04	**Intelligent Enterprise Systems**	Digital transformation and the traits of a modern enterprise architect
05	**AI in Enterprise Systems**	AI-powered ERP, AI decision support, and AI-enabled digital transformation

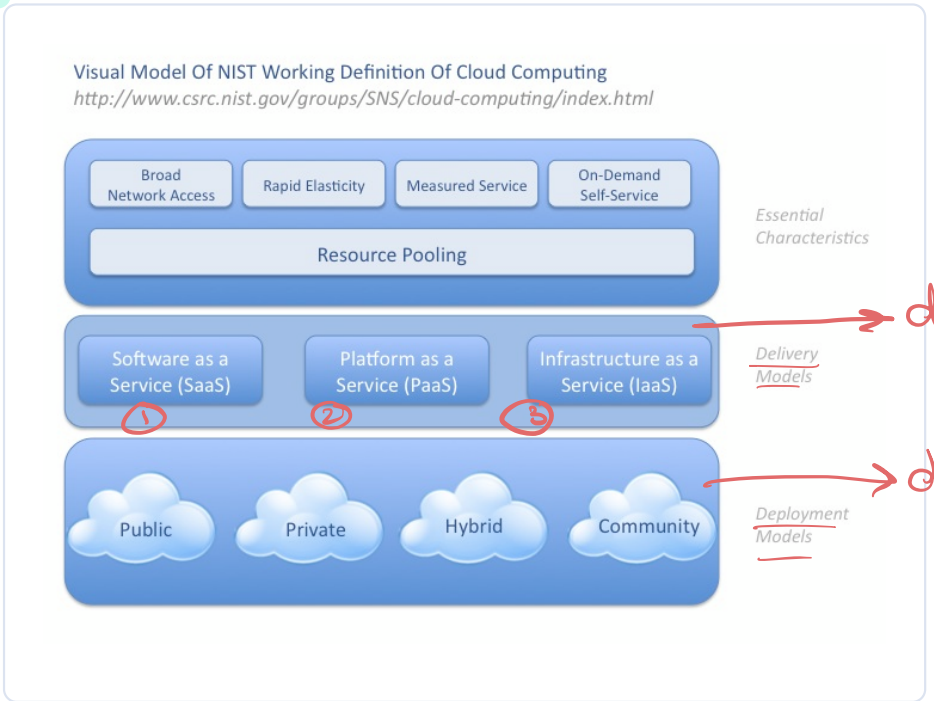
Web Services

A collection of web-based data and procedural resources that use shared protocols and standards, letting different applications share data and services — the foundation of interoperability between systems made by different vendors.



Cloud Computing — The NIST Model

"A model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources… that can be rapidly provisioned and released with minimal management effort." — NIST (Mell & Grance, 2011)



IaaS vs. PaaS vs. SaaS

IaaS

Infrastructure as a Service

Virtualized compute, storage, and networking. You control the OS and apps; the provider controls the physical hardware.

e.g. Amazon EC2, Windows Azure, Google Compute Engine

PaaS

Platform as a Service

A computing platform — OS, language runtime, database, web server. You control your deployed app; the provider controls the platform underneath.

e.g. Google App Engine, Force.com, Windows Azure

SaaS

Software as a Service

Full application access via a thin client (browser or app). You control almost nothing but usage settings; the provider handles everything else.

e.g. Google Apps, Microsoft Office 365, Gmail

Service-Oriented Architecture (SOA)

A business-driven IT architectural approach that integrates a business as linked, repeatable tasks or services — making software components reusable and interoperable via service interfaces, so IT can adapt quickly and economically to changing business needs.

Services

SOA begins with a service — a business task like checking a customer's credit rating. Services are "like" software products, and need to be reusable to move the needle on productivity.

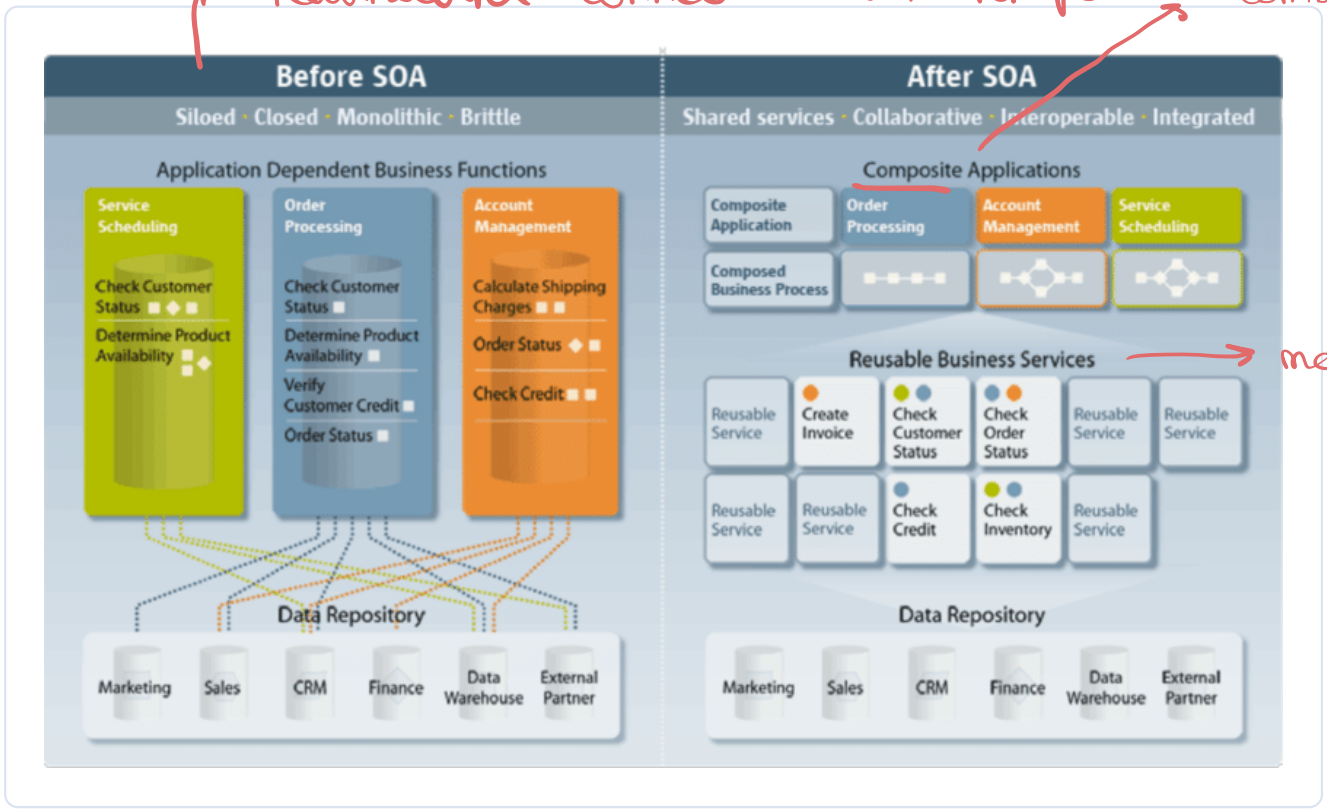
Interoperability

The capability of two or more systems to share data and resources, even when built by different manufacturers — often via a shared format like XML.

Loose Coupling

Services can be joined together on demand to create composite services, or disassembled just as easily into their functional components.

SOA in Practice: Before and After



Virtualization

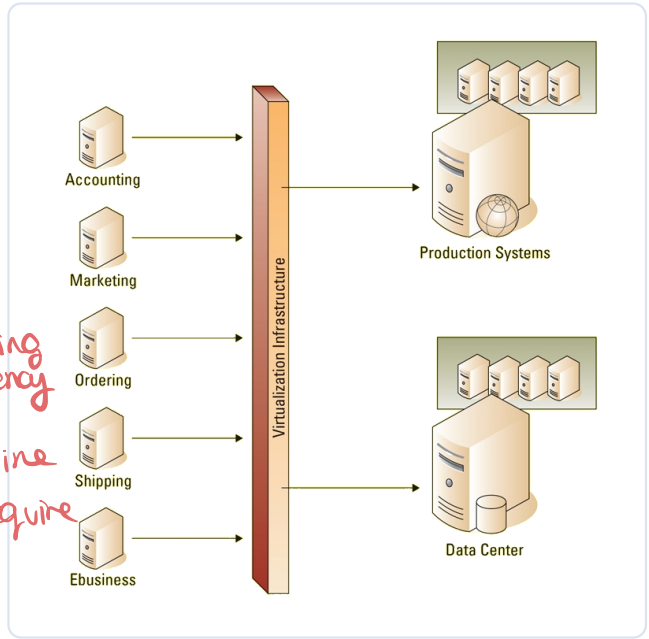
helps done already existing HW

A framework for dividing one computer's resources into multiple execution environments — a way of maximizing the investment already made in hardware, rather than buying more of it.

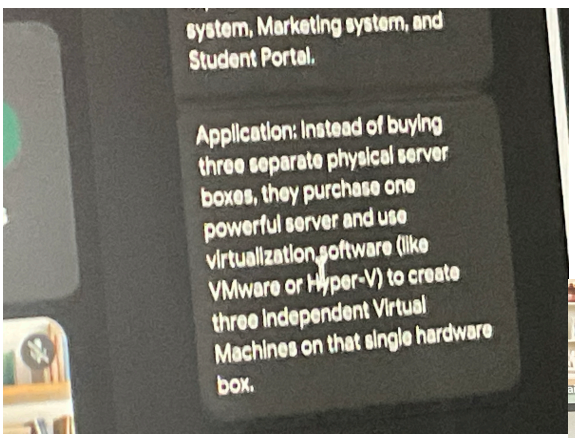
→ cuts down operational costs

Business drivers:

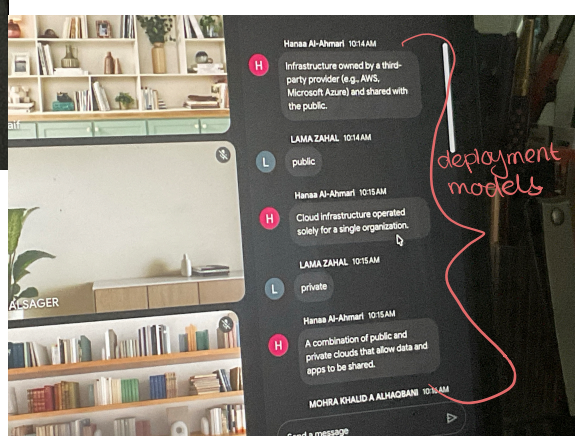
- Underutilized hardware
  - virtualization enable multiple servers to run → maximizing efficiency
  - traditional physical servers cap spans 10-15% -
- Data centers running out of space
  - one physical machine → multiple virtualization machine
- Rising energy costs
  - virtualization saves energy costs ↓ doesn't require cooling equipment + powering the physical server
- Mounting sysadmin costs
  - ↓ cut down IT administration costs

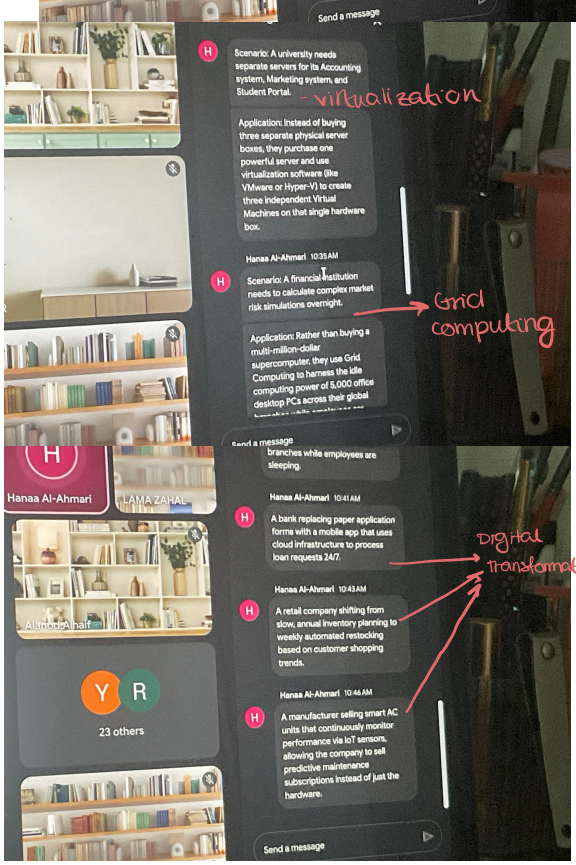


A university needs separate servers for its Accounting, Marketing, and Student Portal system



→ also helps promote scalability at minimal costs if uni wants to expand its system environment



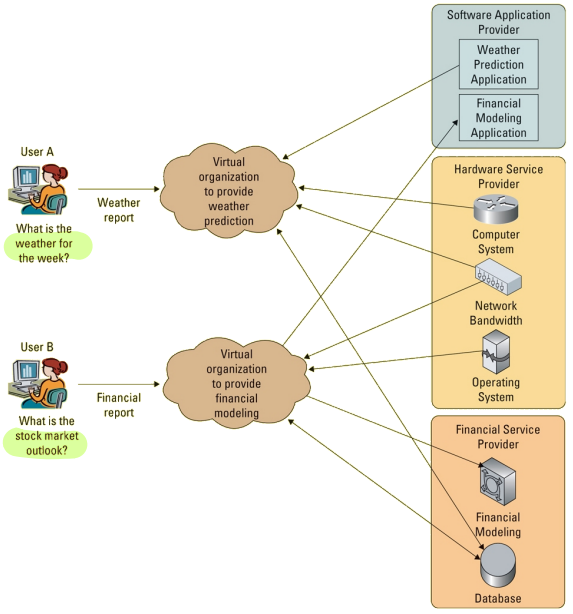


Grid Computing

An aggregation of geographically dispersed computing, storage, and network resources, coordinated to deliver improved performance, better utilization, and easier access to data — letting widely dispersed departments create virtual organizations that share resources.

Business benefits:

- Instantaneous access to massive resources → facilitates easy data access
- Leverages existing capital investments → enhances RoI
- Robust, flexible operational architecture → if one computer fails backups help maintain/uphold operations preventing it from disruptions



NEW SECTION

Intelligent Enterprise Systems

The five technologies above are the plumbing. This section covers how digital transformation — and specifically AI — changes what enterprises build on top of that plumbing.

What Is Digital Transformation?

→ enhances technology base

The profound transformation of business activities, processes, competencies, and models to fully leverage digital change — using digital technologies to transform existing processes and services, or create new ones, to meet evolving market and customer expectations.

leverages modern technologies to transform business labor competencies to meet evolving requirements

Enabled by Real Technologies

Made possible by cloud computing, social, mobile, and analytics — the same technologies covered earlier in this deck.

Pushes Enterprises to Change

Toward being more agile, customer-centric, innovative, efficient, and team-oriented.

Driven by Data and IoT

Big Data and the Internet of Things push organizations toward more service-oriented revenue streams.

↑ physical objects embedded that allow s/w collect + gather data over network

↓ + maximizing productivity

Five Traits of Tomorrow's Enterprise Architect

Digital transformation succeeds or fails partly on whether the EA function evolves. A modern architect needs five key traits:

Ability to Execute

Assesses what's working, grasps and structures information quickly, and stays accessible until a solution is reached.

Tech-Savvy

develop technologies and determine how it fits business landscape

Aware of APIs, microservices, DevOps, and emerging database technologies — and how a proposed service could help or hurt the enterprise.

Agile Mindset

Fluent in Scrum and Kanban; shifts from an "ivory tower" mindset to a project-manager mindset to speed up delivery.

Data-Driven Decisions

"Data will become a strategic asset… enabling the organization to distinguish signals from noise." — Björn Goerke

Evangelistic

must develop skill to voice needed to establish collaboration

Shifts from IT-specific legacy tasks toward facilitating collaboration — Gartner's "Vanguard EA," with a seat close to every C-suite office.

↓ chiefs

AI-Powered ERP Systems



Traditional ERP systems record and route transactions. AI-powered ERP platforms go further — embedding machine learning directly into core business processes to forecast demand, flag anomalies, and automate routine decisions, rather than just logging them for a human to review later.

EXAMPLE

Modern cloud ERP platforms (e.g. SAP S/4HANA Cloud, Oracle Fusion Cloud) now embed AI modules that predict inventory shortfalls before they happen and auto-generate purchase orders — extending the SaaS delivery model covered earlier this week with an intelligence layer on top.

AI in Enterprise Decision Support

The "Data-Driven Decisions" trait of a modern EA (previous slide) describes a mindset. AI-based decision support tools operationalize that mindset — surfacing patterns across the EA repository's business and technology data that a manual review would likely miss, and recommending specific actions rather than just presenting a dashboard.

EXAMPLE

An AI decision-support tool analyzes the EA repository's Week 5 data across all six columns and recommends: "Consolidate these three underused SaaS licenses — combined usage doesn't justify separate contracts."

AI-Enabled Digital Transformation

Digital transformation (this deck's earlier definition) was originally driven by cloud, social, mobile, and analytics. AI is now the newest driver — not replacing those technologies, but making them adaptive: systems that don't just execute a fixed digital process, but continuously learn and adjust it.

→ more dynamic + flexible to adjust processes dynamically to align with evolving requirements e.g. innovation

EXAMPLE

A university's online-degree platform (Weeks 3-4's running example) uses an AI-driven advising chatbot that doesn't just answer static FAQs — it learns from each semester's advising patterns and adjusts its guidance for the next cohort automatically.

CHAPTER SUMMARY

Key Takeaways

- Five technologies support EA's technology-planning link: web services, cloud computing (IaaS/PaaS/SaaS), SOA, virtualization, and grid computing.
- The IaaS/PaaS/SaaS tiers form a control gradient — more control and responsibility at the IaaS end, less (but more convenience) at the SaaS end.
- SOA's before/after picture is the clearest illustration of its value: siloed duplicated logic becomes shared, reusable services.
- Digital transformation succeeds partly on whether the EA function itself evolves — the five traits of a modern architect matter as much as the technology.
- AI extends all three areas covered this week: ERP systems become predictive, decision support becomes prescriptive, and digital transformation becomes adaptive rather than fixed.

COMING UP NEXT

Further Reading & Next Session

Reading Material

- Mell, P., & Grance, T. (2011). The NIST Definition of Cloud Computing
- The Definitive Guide to Digital Transformation with Enterprise Architecture, LeanIX (2019)
- PSU IS321 Syllabus — AI and EA Tools policy

Next: Week 11

The Most Significant EA Frameworks — the Zachman Framework and how it compares to the others we've touched on. Mapped to CLO1, CLO4.

Questions?